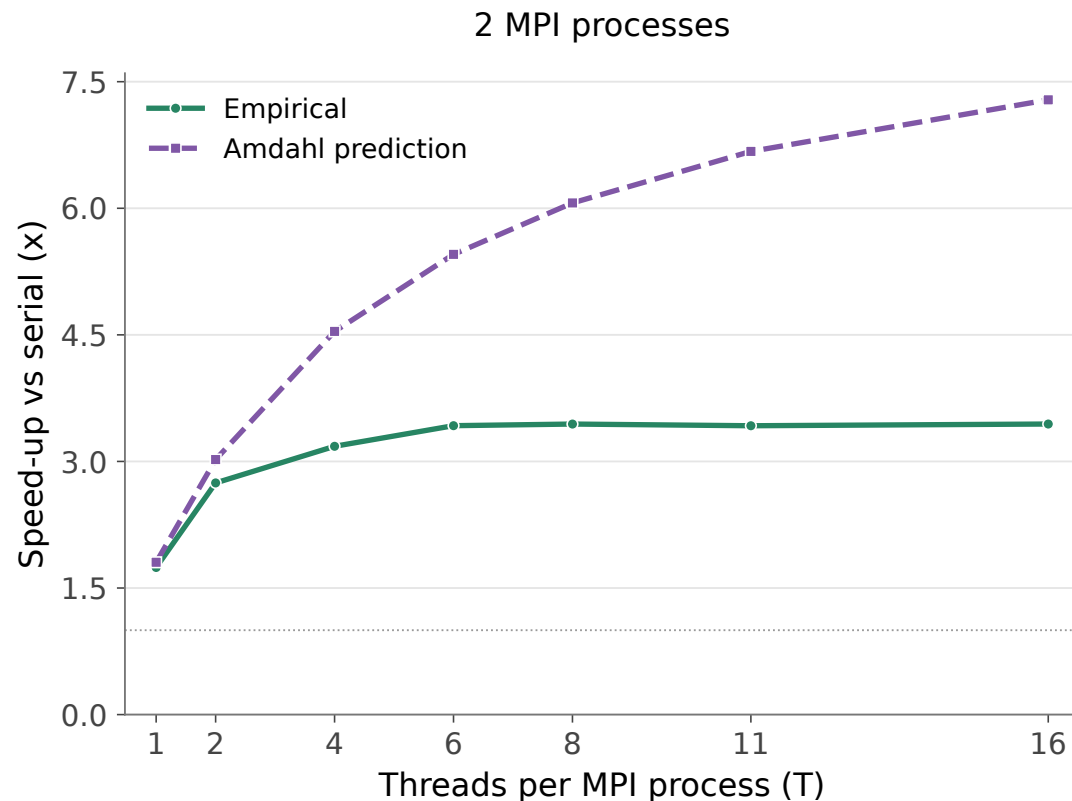
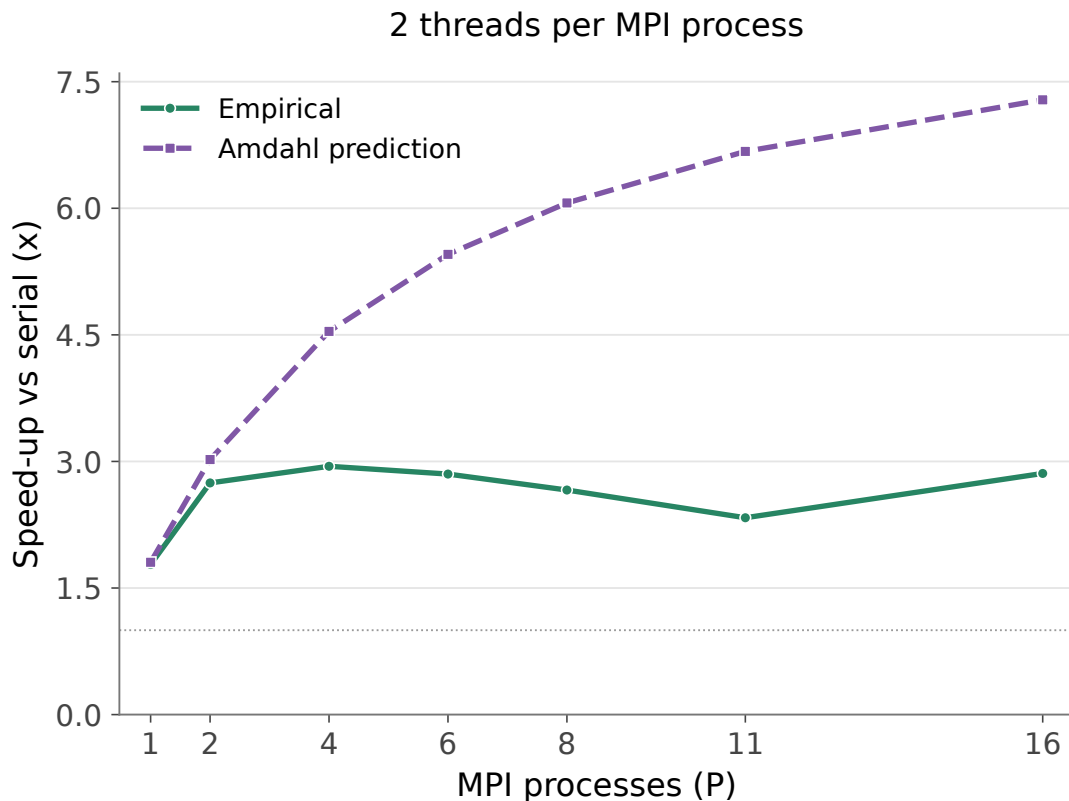


7. Hybrid: empirical vs theoretical speed-up

Amdahl prediction from one MPI process and one OpenMP thread at each n



Each point: median speed-up across 32 n values, 2.5-10 million; median of 3 runs at each n.

Overall wall-clock time: setup + search + gather (MPI) + file output; MPI startup excluded.

Amdahl: $T_{\text{serial}} / (S + C / W)$; S and C measured with one worker; $W = \text{processes} \times \text{threads}$.

Apple M3 Pro | 11 CPU cores | Local measurements